

James Boyko

PERSONAL DETAILS

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University of Michigan
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EDUCATION

Ph.D. Biological Sciences *Aug. 2017 – Aug. 2022*
University of Arkansas, USA
Supervisor: Dr. Jeremy M. Beaulieu
Thesis: *Accounting for rate heterogeneity in trait evolution*

M.Sc. Evolutionary Biology and Ecology *Aug. 2015 – Jul. 2017*
University of Toronto, Canada
Supervisor: Dr. D. Luke Mahler
Thesis: *Character displacement in Lesser Antillean Anolis*

B.Sc. Biology and History *Sept. 2010 – Jun. 2015*
University of Toronto, Canada

PROFESSIONAL EXPERIENCE

Assistant Professor *University of Michigan*
Michigan Society of Fellows *Aug. 2024 – present*

Post-doctoral Researcher *University of Michigan*
Schmidt Futures Post-doctoral Fellow *Jan. 2023 – Dec. 2024*

Post-doctoral Researcher *University of Tennessee, Knoxville*
Post-doctoral Research Associate *Oct. 2022 – Dec. 2022*

PUBLICATIONS

Under Review

Gaem, P. H., **Boyko, J. D.**, Mazine, F. F., Valdemarin, K. S., Smith, S. A., and Vasconcelos, T. (2026). Multiple Trans-Marine Dispersals and Niche Conservatism Underlie the Pantropical Distribution of Eugeniinae (Myrtaceae). *Journal of Biogeography*.

Johnston, I., Diaz-Uriarte, R., and **Boyko, J. D.** (2026). The Hypercubic Mk Model in Reduced State Space for the Coupled, Reversible Coevolution of Multiple Binary Characters. *Systematic Biology*.

Rodrigues Ferreira de Melo, L., Marra, C. C. T., **Boyko, J. D.**, Reginato, M., Morellato, L. P. C., Vasconcelos, T., and de Brito, V. L. G. (2026). Evolutionary Correlation of Floral and Fruit Traits Reflects Trade-offs Between Pollination and Seed Dispersal in Melastomataceae. *Perspectives in Plant Ecology, Evolution and Systematics*.

- Boyko, J. D.** (2026). Automatic Discovery of Optimal Discrete Character Models. *Systematic Biology*, syag018.
- Boyko, J. D.**, Gontjes, K. J., Snitkin, E. S., and Smith, S. A. (2026). Resolving competing evolutionary histories in joint ancestral state reconstruction. *Proceedings of the National Academy of Sciences* 123.15, e2528064123.
- Boyko, J. D.** and Rabosky, D. L. (2026). The Geometry of Macroevolution: Phenotypic Evolution on Non-Euclidean Manifolds. *The American Naturalist* 0.0, pp. 000–000.
- Boyko, J. D.** (2025). SegmentR: Deep Learning for Automated Segmentation with an R Interface. *Ecological Informatics*, p. 103259.
- Gontjes, K. J., Singh, A., Sansom, S. E., **Boyko, J. D.**, Smith, S. A., Lautenbach, E., and Snitkin, E. (2025). Phylogenetic Context of Antibiotic Resistance Provides Insights into the Dynamics of Resistance Emergence and Spread. *The Journal of Infectious Diseases*, jiaf478.
- Stepanova, N., **Boyko, J. D.**, Lin, J., Davis Rabosky, A. R., and Rabosky, D. L. (2025). Punctuated Versus Gradual Shifts in the Multivariate Evolutionary Process: A Test With Paired Radiations of Scincid Lizards. *Systematic Biology* 74.3, pp. 483–498.
- Vasconcelos, T. and **Boyko, J. D.** (2025). mvh: An R Tool to Assemble and Organize Virtual Herbaria from Openly Available Specimen Images. *Applications in Plant Sciences* 13.2, e11631.
- Vasconcelos, T., Weaver, W. N., Baumgartner, A., Bugnaski, Z., and **Boyko, J. D.** (2025). Automated Extraction of Leaf Mass per Area from Digitized Herbarium Specimens. *New Phytologist*.
- Boyko, J. D.** and O'Meara, B. C. (2024). dentist: Quantifying Uncertainty by Sampling Points Around Maximum Likelihood Estimates. *Methods in Ecology and Evolution* 15.4, pp. 628–638.
- Hagen, E. R., Vasconcelos, T., **Boyko, J. D.**, and Beaulieu, J. M. (2024). Investigating Historical Drivers of Latitudinal Gradients in Polyploid Plant Biogeography: A Multiclade Perspective. *American Journal of Botany*, e16356.
- Boyko, J. D.**, Hagen, E. R., Beaulieu, J. M., and Vasconcelos, T. (2023). The Evolutionary Responses of Life-History Strategies to Climatic Variability in Flowering Plants. *New Phytologist*.
- Boyko, J. D.**, O'Meara, B. C., and Beaulieu, J. M. (2023). A Novel Method for Jointly Modeling the Evolution of Discrete and Continuous Traits. *Evolution*.
- Boyko, J. D.** and Beaulieu, J. M. (2022). Reducing the Biases in False Correlations Between Discrete Characters. *Systematic Biology*.
- Mortimer, S. M. E., **Boyko, J. D.**, Beaulieu, J. M., and Tank, D. C. (2022). Synthesizing Existing Phylogenetic Data to Advance Phylogenetic Research in Orobanchaceae. *Systematic Botany* 47, pp. 533–544.
- Boyko, J. D.** and Beaulieu, J. M. (2021). Generalized Hidden Markov Models for Phylogenetic Comparative Datasets. *Methods in Ecology and Evolution* 12.3, pp. 468–478.
- Vasconcelos, T., **Boyko, J. D.**, and Beaulieu, J. M. (2021). Linking Mode of Seed Dispersal and Climatic Niche Evolution in Flowering Plants. *Journal of Biogeography* 50, pp. 43–56.
- Nakov, T., **Boyko, J. D.**, Alverson, A. J., and Beaulieu, J. M. (2017). Models with Unequal Transition Rates Favor Marine Origins of Cyanobacteria and Photosynthetic Eukaryotes. *Proceedings of the National Academy of Sciences* 114, E10606–E10607.

Preprints

- Boyko, J. D.**, Cohen, J., Fox, N., Han Veiga, M., Li, J. I.-H., Liu, J., Modenesi, B., Rauch, A. H., Reid, K. N., Tribedi, S., Vishner, A., and Xie, X. (2023). An Interdisciplinary Outlook on Large Language Models for Scientific Research. arXiv:2311.04929 [cs.CL].

AWARDS AND GRANTS

UM/OpenAI Collaboration Grant
\$50,000 USD

2025

Journal of Biogeography Innovation Award

2022

\$750 USD

Outstanding Graduate Student Research Award
\$1,000 USD

2021

Distinguished Doctoral Fellowship
\$80,000 USD

2017–2021

PROFESSIONAL SERVICE

SSE Symposium — AI in Evolutionary Biology (organizer) 2025

Workshop — AI for Freshwater Biodiversity Conservation (participant) 2025

BioBlend Workshop: AI-Driven Biodiversity Monitoring (organizer) 2024

MIDAS and Schmidt Futures AI in Science Colloquium (organizer) 2023

CODING AND HIGH PERFORMANCE COMPUTING

R	Advanced
High Performance Computing	Advanced
Deep Learning & Computer Vision	Advanced
Large Language Models	Advanced
Python	Proficient
Bash	Proficient

SOFTWARE CONTRIBUTIONS

<i>phyloAMR</i>	Phylogenetically-informed analysis of antibiotic resistance
<i>SegmentR</i>	R package for deep-learning image segmentation
<i>mvh</i>	Assembling and organizing virtual herbaria
<i>corHMM</i>	Correlated Hidden Markov Models for comparative biology
<i>dentist</i>	Uncertainty quantification around maximum likelihood estimates
<i>hiSSE</i>	Hidden State Dependent Speciation Extinction models
<i>OUwie</i>	Continuous character evolution under Ornstein-Uhlenbeck models

COMPETITIONS AND DATA CHALLENGES

Japan Aerospace Exploration Agency (JAXA) 2023
8th place

PEER REVIEW

American Journal of Botany, American Naturalist, Evolution, Frontiers in Artificial Intelligence, Journal of Biogeography, Methods in Ecology and Evolution, Nature Communications, New Phytologist, Pest Management Science, PLOS Biology, PLOS Computational Biology, PNAS, Proceedings of the Royal Society B, Systematic Biology

TEACHING

University of Michigan

- Instructor — BIO202: Biological Data Analysis and Programming (FA 2025)

University of Arkansas

- Teaching Assistant — Evolutionary Biology (2017–2022)

University of Toronto

- Teaching Assistant — Macroevolution (2016–2017)
- Teaching Assistant — Biostatistics (2016–2017)
- Teaching Assistant — From Genomes to Ecosystems in a Changing World (2015–2016)
- Teaching Assistant — Adaptation and Biodiversity (2015–2016)

Workshops and Short Courses

- Instructor — AI for Scientists and Engineers Summer Academy: Foundation Models, UM (2025).
- Instructor — Workshop Harmonizing Accumulation Modeling (WHAM), University of Bergen (2024).
- Instructor — Gans Foundation Student Workshop: Functional Morphology, UM (2024)
- Instructor — Phylogenetic Comparative Methods, Stats Workshop, UM (2024)
- Instructor — Phylogenetic Comparative Methods in R, Boston University (2024)

PRESENTATIONS

Invited Talks

1. Boyko, J.D. (2026) Phylogenetic inference of variation in the propensity for trait emergence. *MIDGE Annual Meeting*.
2. Boyko, J.D. (2026) Generative AI for Literature Reviews. *MIDAS Generative AI Tutorial Series*.
3. Boyko, J.D. (2026) Resolving competing evolutionary histories in joint ancestral state reconstruction. *NIAID GCID Annual Meeting*.
4. Boyko, J.D. (2025) HMMs and antibiotic resistance evolution. *DCMB Tools and Technology Seminar*.
5. Boyko, J.D. (2025) An introduction to Foundation Models (Large Language Models). *MIDAS Generative AI Tutorial Series*.
6. Boyko, J.D. (2024) Writing, Planning, and Literature Review: Enhancing Professional Productivity with Generative AI. *MIDAS Generative AI Tutorial Series*.
7. Boyko, J.D. (2023) An introduction to generative AI tools for research. *MIDAS Generative AI Tutorial Series*.

Contributed Presentations

1. Boyko, J.D. (2025) Automatic discovery of optimal discrete character models. *Evolution* — Oral, International.
2. Boyko, J.D., Vasconcelos, T.N.C. (2024) Rates of biome shift predict diversification dynamics in flowering plants. *International Botanical Congress* — Oral, International.
3. Cohen, Y., Rauch, A., Li, J., Modenesi, B., Boyko, J.D., Wang, Y., Byon, E., Huan, X. (2023) Sparse modeling of wavelet features for fault classification in spacecraft propulsion systems. *PHMAP* — Poster, International.

4. Boyko, J.D., Hagen, E.R., Beaulieu, J.M., Vasconcelos, T.N.C. (2023) Evolutionary responses of life-history strategies to climatic variability. *Botany* — Oral, International.
5. Boyko, J.D., O'Meara, B.C., Beaulieu, J.M. (2023) A novel method for jointly modeling discrete and continuous trait evolution. *Evolution* — Oral, International.
6. Boyko, J.D., Beaulieu, J.M. (2022) Reducing the biases in false correlations between discrete characters. *Evolution* — Poster, International.
7. Boyko, J.D., Beaulieu, J.M. (2019) Quantifying the limits of our knowledge in phylogenetic comparative studies. *Quantitative Genetics Workshop* — Oral, International.
8. Boyko, J.D., Nakov, T., Alverson, A.J., Beaulieu, J.M. (2019) Quantifying the limits of our knowledge in phylogenetic comparative studies. *Botany* — Oral, International.
9. Boyko, J.D., Nakov, T., Alverson, A.J., Beaulieu, J.M. (2018) Testing the signal of marine vs. freshwater origins of photosynthetic eukaryotes. *Botany* — Oral, International.
10. Boyko, J.D., Mahler, D.L. (2017) Testing for character displacement in Lesser Antillean anoles. *Atwood Colloquium* — Oral, Institutional.
11. Boyko, J.D., Mahler, D.L. (2016) Character displacement in Lesser Antillean *Anolis* lizards. *Evolution* — Oral, International.
12. Boyko, J.D., Mahler, D.L. (2016) Testing for repeated character displacement in *Anolis* lizards. *OE3C* — Oral, Provincial.

REFERENCES

Dr. Jeremy M. Beaulieu

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